Pakala Tarun Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.33
Sri Chaitanya	Class XII (SSC)	2022	9.70%
Sri Chaitanya	Class X (TSBIE)	2024	98.90%

RESEARCH EXPERIENCE

Undergraduate Researcher -- Machine Unlearning | Dept. of AI, IIT Hyderabad (Advisor: Prof. Konda Reddy Mopuri) -- *May 2026-Present*

- Researching **machine unlearning** techniques under faculty guidance to remove sensitive data or harmful learned concepts from trained models **without full retraining**, benchmarking against gold-standard retrained-from-scratch performance
- Conducting a structured literature review across **generative and discriminative unlearning** approaches to identify gaps and inform an original methodology
- Building foundational understanding of unlearning evaluation metrics (**membership inference resistance**, forgetting efficacy vs. retained accuracy trade-offs)

PROJECTS

ML Framework Development (Mini-PyTorch & Mini-Scikit-learn) | Python, NumPy

- Built a modular Python ML library (my_ml_lib) from scratch, implementing a **scikit-learn-style API** across preprocessing, linear models, and model-selection modules
- Designed a **custom autograd engine** (reverse-mode automatic differentiation) supporting matrix multiplication, activations, and gradient propagation through dynamically built computation graphs
- Implemented Logistic Regression via **Newton-Raphson/IRLS** and a from-scratch neural network stack (Linear, ReLU, Sigmoid, SGD, CrossEntropyLoss) on the custom autograd engine
- Benchmarked **5 architectures** on **Fashion-MNIST**, tuning hyperparameters via **k-fold cross-validation**

Image Classification Hackathon -- CNN-BiLSTM Hybrid | PyTorch -- *Team "HailMary," Deep Learning Course, IIT Hyderabad*

- Designed and trained the team's best-performing model: a dual-branch architecture fusing a **ResNet-style CNN backbone with Squeeze-and-Excitation blocks** and a **bidirectional 2-layer LSTM with attention pooling** over patch-embedded sequences, achieving **75.79% accuracy** and a **top-25% finish among 74 competing teams**
- Trained from scratch (no pretrained backbones, per competition rules) using **AdamW with cosine-annealed warmup**, **mixed-precision training**, and **gradient clipping** for stable convergence over 35 epochs
- Applied **MixUp/CutMix augmentation with label smoothing** during training and **8-view test-time augmentation** at inference to close the gap between validation and leaderboard accuracy
- Owned the full ML pipeline end-to-end -- data loading, preprocessing, model architecture, training loop, and final submission generation

Deep Learning From Scratch | Python, NumPy, PyTorch

- Implemented core deep learning architectures from first principles: **MLPs with backpropagation**, **CNNs** (custom convolution/pooling), **RNNs/LSTMs with BPTT** and gradient clipping, and a **Transformer encoder** with multi-head self-attention
- Built and compared sequence models (LSTM, Transformer, **Linear/Selective State Space Models**) on long-context sentiment classification, analyzing computational scaling and gradient propagation trade-offs

SKILLS

Languages: Python, C, SQL, LaTeX

Frameworks/Libraries: PyTorch, Scikit-learn, NumPy, Matplotlib, Pandas

RELEVANT COURSES

- **AI:** Foundations of Machine Learning, Convex Optimization I & II, **Deep Learning**, AI for Humanity
- **Mathematics:** Probability & Random Variables, Vector Calculus, Differential Equations, Calculus I & II
- **Computer Science: Data Structures, Algorithms**, DBMS, Operating Systems(Intoductory)
- **Electrical Engineering:** Control Systems, Linear Systems & Signal Processing, Matrix Theory